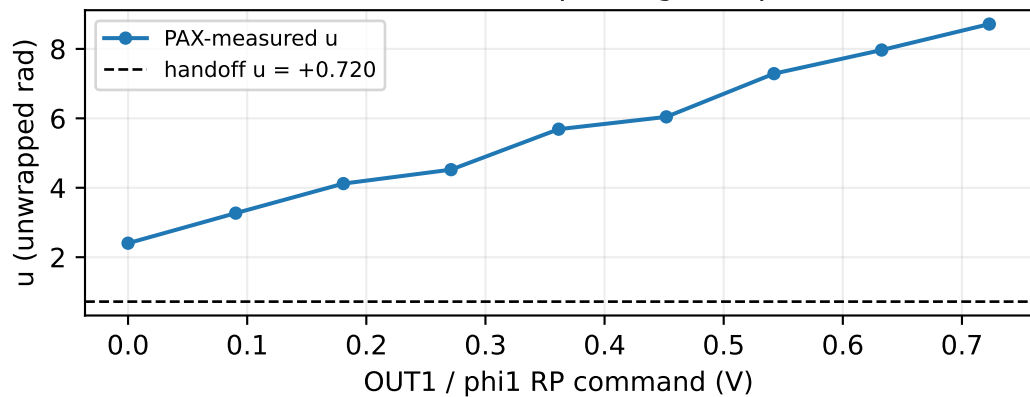
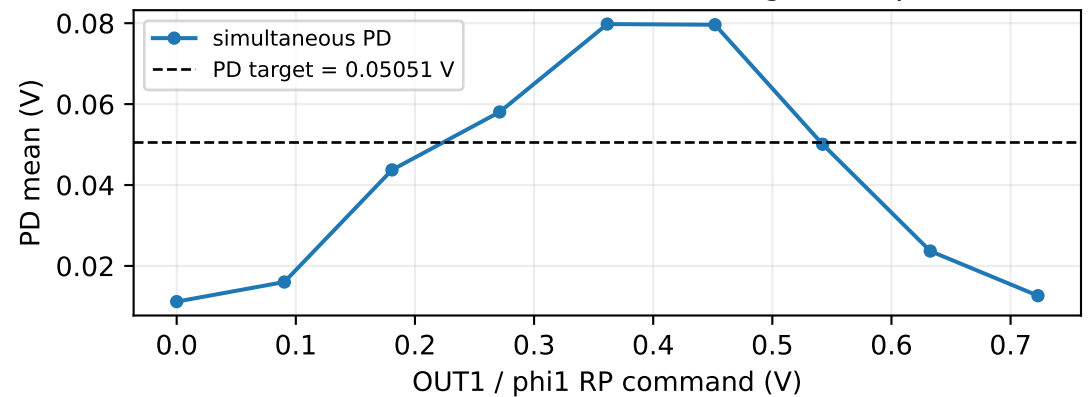


Phi1 PD FPGA lock — 010148_phi1-pd-lock-test_phi1-pd-lock-5

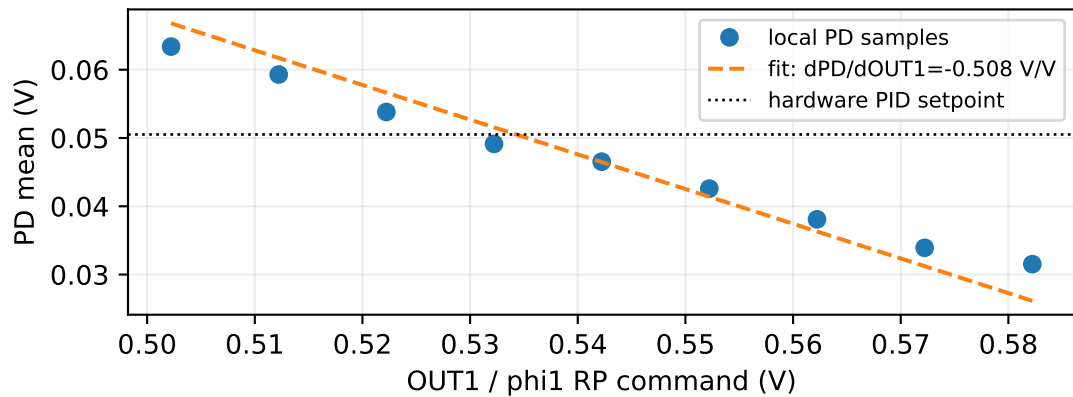
PAX-assisted $\phi 1$ rough map



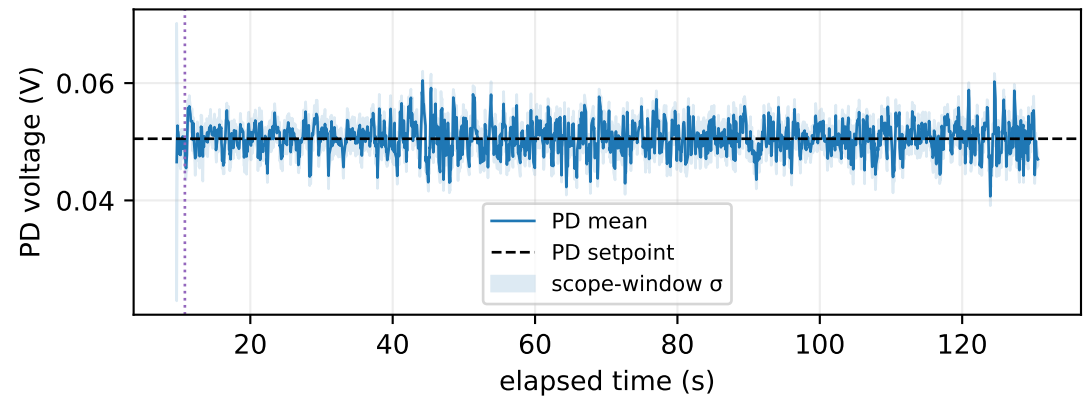
PD value associated with rough u map



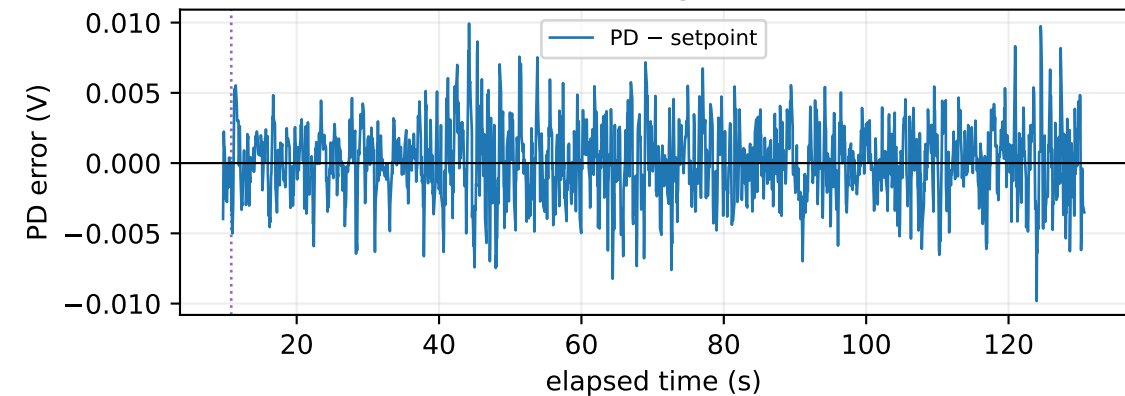
Automatic local PD calibration



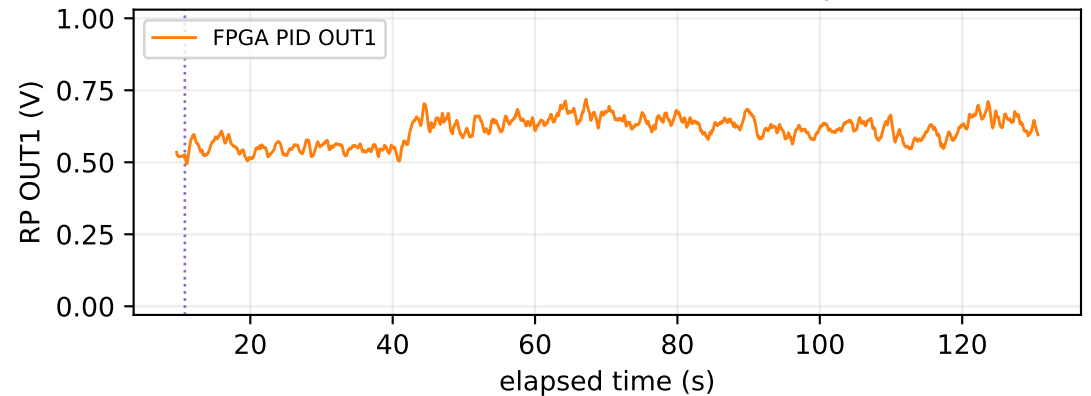
FPGA PD lock



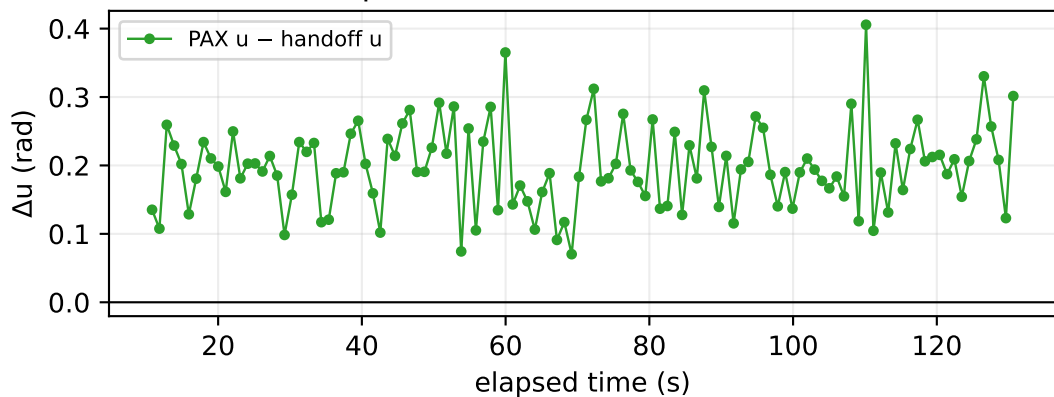
PD locking error



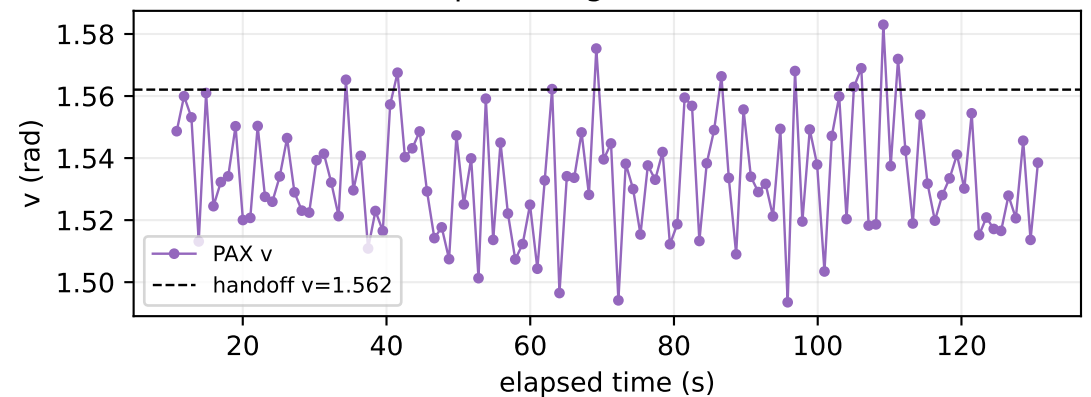
Continuous FPGA actuator output



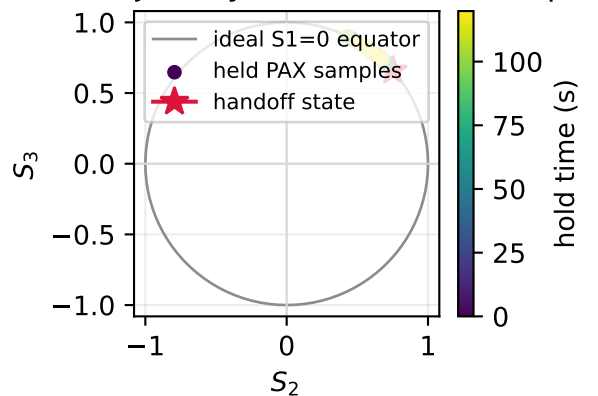
Independent PAX azimuth validation



PAX polar-angle cross-check



Held lock trajectory on the Poincare equator



Interpretation

- Tight cluster near the red star: PD lock preserves the selected u branch.
- Motion around the circle: residual $\phi 1/u$ drift or branch movement.
- Radial motion: polarization is leaving the ideal equator.